a. Radio Iodine b. Radio Cobalt

c. Radio Carbon d. Radio Nickel

6. Gamma radiations are dangerous because

a. it affects eyes & bones

b. it affects tissues

c. it produces genetic disorder

d. it produces enormous amount of heat

7. _____ aprons are used to protect us from gamma radiations

a. Lead oxide b. Iron

c. Lead d. Aluminium

8. Which of the following statements is/are correct?

i. α particles are photons

ii. Penetrating power of γ radiation is very low

iii. Ionization power is maximum for α rays

iv. Penetrating power of γ radiation is very high

a. (i) & (ii) are correct

b. (ii) & (iii) are correct

c. (iv) only correct

d. (iii) & (iv) are correct

9. Proton - Proton chain reaction is an example of _____

a. Nuclear fission b. α - decay

c. Nuclear fusion d. β - decay

10. In the nuclear reaction ${}_{6}X^{12} \xrightarrow{\alpha \text{ decay}} {}_{Z}Y^{A}$, the value of A & Z.

a. 8, 6 b. 8, 4

c. 4, 8 d. cannot be determined with the given data

11. Kamini reactor is located at _____

a. Kalpakkam b. Koodankulam

c. Mumbai d. Rajasthan

12. Which of the following is/are correct?

i. Chain reaction takes place in a nuclear reactor and an atomic bomb.

ii. The chain reaction in a nuclear reactor is controlled

iii. The chain reaction in a nuclear reactor is not controlled

iv. No chain reaction takes place in an atom bomb

a. (i) only correct b. (i) & (ii) are correct

c. (iv) only correct d. (iii) & (iv) are correct

II. Fill in the blanks

1. One roentgen is equal to _____ disintegrations per second
2. Positron is an_____.
3. Anemia can be cured by _____ isotope
4. Abbreviation of ICRP_____
5. _____is used to measure exposure rate of radiation in humans.
6. _____ has the greatest penetration power.
7. ${}_{Z}Y^{A} \rightarrow {}_{Z+1}Y^{A} + X$; Then, X is _____
8. ${}_{Z}X^{A} \rightarrow {}_{Z}Y^{A}$ This reaction is possible in _____ decay.
9. The average energy released in each fusion reaction is about _____ J.
10. Nuclear fusion is possible only at an extremely high temperature of the order of _____ K.
11. The radio isotope of _____ helps to increase the productivity of crops.
12. If the radiation exposure is 100 R, it may cause _____.

III State whether the following statements are true or false: If false, correct the statement

1. Plutonium -239 is a fissionable material.
2. Elements having atomic number greater than 83 can undergo nuclear fusion.
3. Nuclear fusion is more dangerous than nuclear fission.
4. Natural uranium U-238 is the core fuel used in a nuclear reactor.
5. If a moderator is not present, then a nuclear reactor will behave as an atom bomb.
6. During one nuclear fission on an average, 2 to 3 neutrons are produced.
7. Einstein's theory of mass energy equivalence is used in nuclear fission and fusion.

10th_Science_Unit-6.indd 87 11-01-2020 10:26:52